

Niraj Amrutkar

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PROFESSIONAL SUMMARY

Software Engineer experienced in building scalable backend systems, distributed data pipelines and production-grade AI applications using **Java, Python, Spring Boot, FastAPI, Kafka and OpenSearch**. Built REST APIs, microservices, event-driven systems and high-throughput document processing pipelines handling **30M+ documents**. Experienced in **RAG, hybrid search, self-hosted embeddings, GPU acceleration**, with strong foundations in distributed systems, system design, DSA and GenAI.

TECHNICAL SKILLS

Programming	Java, Python, C/C++
Frameworks	Spring Boot, FastAPI, REST APIs, Microservices, gRPC, ReactJs
Distributed Systems	Apache Kafka, Event-Driven Architecture, Distributed Processing
AI GenAI	LLMs, RAG, LangChain, LangGraph, MCP, Reranking
Search & Data	OpenSearch, Elasticsearch, BM25, Semantic Search, PostgreSQL, NoSQL
Cloud & Infrastructure	Docker, Kubernetes, Helm, AWS, Prometheus, Confluent Kafka
Engineering Fundamentals	Data Structures & Algorithms, OOP, System Design, Design Patterns, Git

WORK EXPERIENCE

S&P Global <i>Associate Software Engineer</i>	July 2024 – Present <i>Hyderabad, India</i>
- Developed production-grade Java Spring Boot backend services with efficient service-to-service communication using gRPC. - Modernized Java applications by migrating Java 17 to Java 21 and Gradle to Maven, ensuring compatibility and standardized builds. - Designed and deployed a production-grade RAG system for processing 40–50 GB of enterprise data using Python, FastAPI and OpenSearch. - Built scalable bulk ingestion pipelines for text extraction, chunking, embedding generation and indexing, processing 30M+ documents through distributed Kafka workflows. - Hosted embedding models on dedicated servers for private-cloud inference, eliminating external API dependencies and keeping sensitive data within private infrastructure. - Optimized embedding generation using GPU acceleration, asynchronous processing and batching to maximize GPU utilization and improve processing throughput. - Engineered semantic and hybrid retrieval using OpenSearch, combining BM25 and dense embeddings and improving top-k retrieval precision by 15%+. - Designed and deployed scalable containerized microservices using Docker, Kubernetes and Helm for highly available production deployments.	

PROJECTS

Hybrid Search RAG System

LangChain, OpenSearch, FastAPI, Ollama

Architected an end-to-end **RAG pipeline** covering ingestion, chunking, embeddings, vector indexing, retrieval, reranking and LLM-based answer synthesis.

Implemented **hybrid semantic + BM25 search** with cross-encoder reranking; developed FastAPI APIs and evaluated retrieval using **precision@k** to improve answer grounding.

EDUCATION

Vishwakarma Institute of Information Technology B.Tech in Information Technology, CGPA: 9.27/10 <i>Coursework: Data Structures, Algorithms, DBMS, Operating Systems, Probability, System Design</i>	Nov 2020 – May 2024 Pune, India
GATE (CS-IT) 2024 — TCS CodeVita 2023 – Top 0.14%	